



Iowa City Robotics

Catapults

Summary

Students build a simple catapult and launch marshmallows to explore how pushing, pulling, and stored energy make things move.

Description

Students work together to build a small catapult using popsicle sticks, a spoon, and a rubber band. When they pull the spoon back and let go, the marshmallow launches into the air. Students can test how far their marshmallow travels and try small changes to improve their catapult.

Materials

Material	Quantity
Mini marshmallows	Several per group
Plastic spoons	1 per group
Popsicle sticks	10–12 per group
Tape (duct or masking)	Small amount per group
Rubber bands	1–2 per group
Ruler or measuring tape	1 per classroom
Bucket or target (optional)	1 per classroom

Directions

- Divide students into small groups (2–3 students)
- Give each group popsicle sticks, a spoon, tape, and a rubber band
- Help students build a small base using popsicle sticks and tape
- Stretch the rubber band between two sticks at the front of the catapult
- Place the spoon in front of the rubber band
- To launch
 - Pull the spoon back
 - Place a marshmallow in the spoon



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- Let go and watch it launch
- Measure how far the marshmallow travels
- Let students try again and see if they can launch it farther

Tips

- One student can hold the base, so the catapult does not move
- Pull the spoon back before placing the marshmallow so it does not fall out
- Make sure students launch marshmallows away from others

Activity Extension

- Try launching from different pull-back distances
- Try different rubber bands and see which launches farther
- Create a target or bucket and see who can land the marshmallow inside

Discussion Questions

- What happened when you pulled the spoon back?
- What happened when you let go?
- Did the marshmallow go far or short?
- What could you change to make it go farther?
- How did your team work together to build the catapult?

What is Happening?

- When the spoon is pulled back, the rubber band stretches and stores energy
- When the spoon is released, that energy launches the marshmallow
- Pulling back farther stores more energy and can make the marshmallow travel farther
- The popsicle stick frame helps keep the catapult stable while launching